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Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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VII SEMESTER

Experiential Learning - ISM

Phase 1

on

DRIVE: Data-Driven Reliability Intelligence for Vulnerability Estimation

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Bengaluru - 560059, Karnataka, India

Team Members

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SL NO	NAME	USN
1	Sumedh Udupa Uppinkudru	1RV23CS250
2	Syed Muzammil Hussaini	1RV23CS263
3	Varun Gowda R	1RV23CS281

MENTOR

Dr. Prapulla S B
Associate professor
Department of CSE
RV College of Engineering
Bengaluru-59

RV College of Engineerir
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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Introduction

Managing Reliability Across Large SSD Fleets

- Modern data centres rely on large SSD populations to support cloud services, databases and critical applications.
- Drive failures can lead to service disruption, data unavailability and additional maintenance costs.
- Failure occurrences may vary across SSD models and infrastructure locations such as racks and nodes.
- Since inspecting or replacing every drive is impractical, maintenance teams must prioritise where resources are allocated.

Project Motivation

Use historical SSD failure patterns to identify drives and infrastructure groups that deserve maintenance attention first.

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Literature Review

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S.No	Title of Paper	Author(s)	Key Findings	Research Gaps (Considering Our Project)
1	An In-Depth Study of Correlated Failures in Production SSD-Based Data Centers (2021)	Shujie Han et al.	Analyses large-scale SSD failure records and identifies correlated failures across infrastructure locations.	Does not compare budget-constrained maintenance selection strategies using model-wise and infrastructure-wise failure patterns.
2	Prophet: SSD Failure Analysis and Prediction Guided by Flash Reliability Characteristics in Data Centers (2025)	Yunpeng Song, Yujiong Liang, Jialin Liu, Liang Shi	Incorporates flash-memory reliability characteristics into SSD failure analysis and prediction.	Does not focus on converting historical failure patterns into explainable maintenance priorities.
3	Temporal-Contextual Attention Network for Solid-State Drive Failure Prediction in Data Centers (2024)	Chankyu Koh, Jisu Kang, Taehyeong Kim, Sung Won Han	Uses LSTM and Transformer-based attention to capture temporal patterns and relationships among SMART attributes.	Focuses on prediction accuracy rather than resource-constrained maintenance selection and prioritisation.

Literature Review

S.No	Title of Paper	Author(s)	Key Findings	Research Gaps (Considering Our Project)
4	Multiple Instance Learning for SSD Failure Prediction Under Customer Failure-Biased Labels (2026)	Bongjun Choi, Jeongwon Park, Hyung-Seok Kang, Jaewoong Shim	Uses multiple instance learning and spatial-temporal information to improve prediction under noisy failure labels.	Does not investigate comparative maintenance prioritisation strategies under fixed resource budgets.
5	Estimating Mean Time to Failure of Solid-State Drives via Self-Organizing Map and Model Averaging (2025)	Peng Li, Xun Xiao, Jiayu Chen	Combines health-indicator construction and degradation modelling to estimate SSD mean time to failure.	Does not address fleet-level maintenance prioritisation using infrastructure-level failure patterns.
6	Online Maintenance Prioritization via Monte Carlo Tree Search and Case-Based Reasoning (2022)	Michael Hoffman, Eunhye Song, Michael Brundage, Soundar Kumara	Uses Monte Carlo Tree Search and case-based reasoning to prioritise maintenance under resource constraints.	Focuses on general maintenance decision-making rather than SSD-specific historical failure analysis and infrastructure-aware selection.

Problem Definition

Data centre operators manage large SSD populations with limited maintenance resources. Historical failure patterns vary across SSD models and infrastructure groups, making uniform or random maintenance selection potentially inefficient.

A systematic approach is required to identify and prioritise SSDs for maintenance while maximising the coverage of historically observed failures.

Objectives

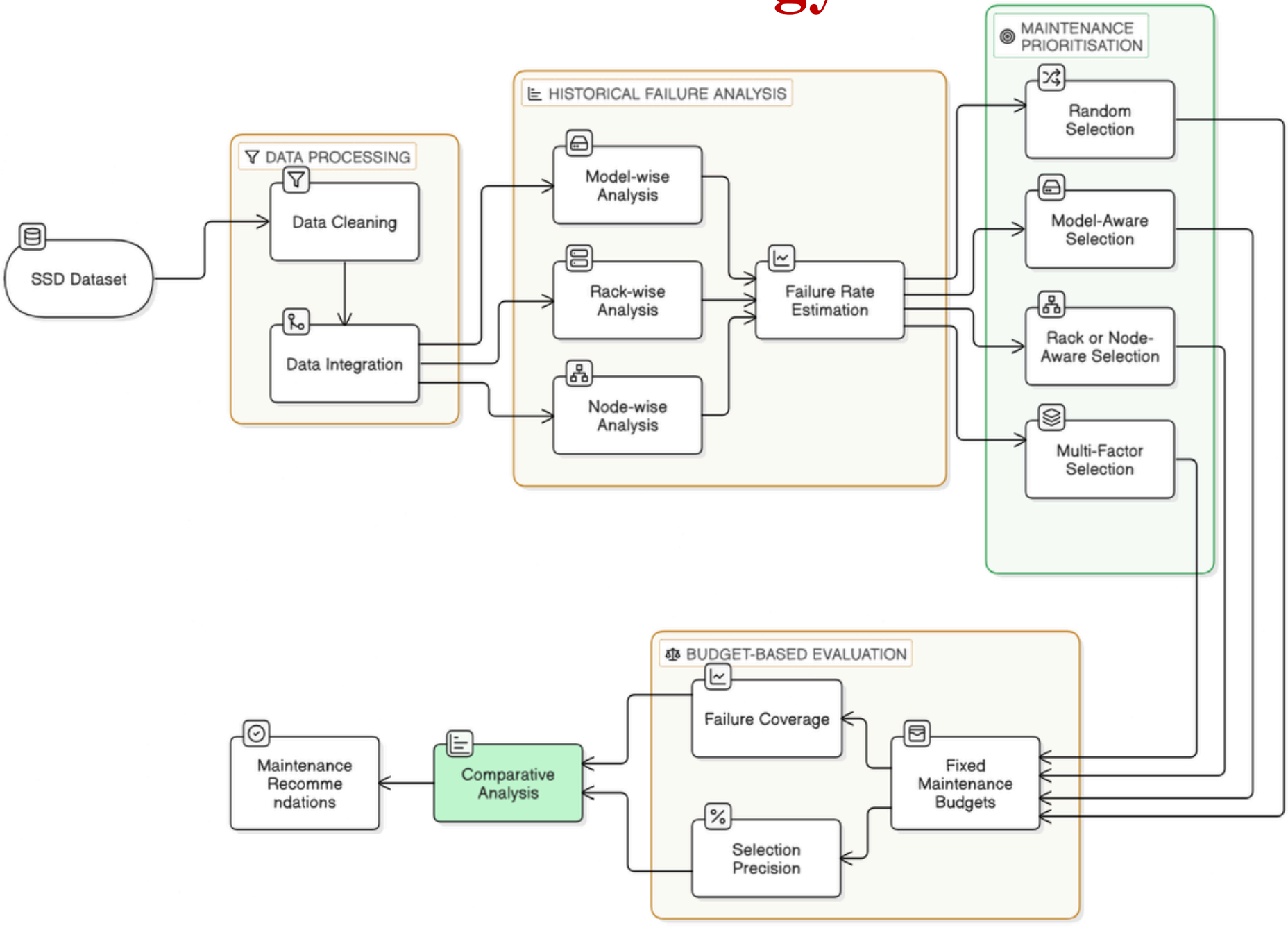
Primary Objectives:

- To analyse historical SSD failure patterns across SSD models and infrastructure groups.
- To develop explainable, failure-aware maintenance prioritisation strategies using historical failure distributions.
- To evaluate and compare maintenance strategies under different maintenance budget constraints.

Secondary Objectives:

- To visualise SSD failure patterns, maintenance priorities and strategy performance through an interactive dashboard.
- To provide interpretable recommendations for efficient allocation of limited maintenance resources.

Methodology



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Dataset Overview

PROPOSED DATA SOURCE — BACKBLAZE DRIVE STATS

Dataset overview

Dataset at a glance

- Quarterly drive statistics
- Daily drive-level observations
- SSD model information
- Failure indicators
- SMART attributes
- Infrastructure identifiers

Key attributes

- `date`
- `serial_number`
- `model`
- `failure`
- SMART fields
- `pod_id`, `vault_id`
- `cluster_id`, `datacenter`

Relevance to the project

Dataset information	Analysis
SSD model and failure indicator	Model-wise failure analysis
Pod, vault and cluster identifiers	Infrastructure-wise analysis
SMART attributes	Drive health characteristics
Date and serial number	Historical drive-level tracking

Data source: Backblaze Drive Stats – official dataset

Note: SSD-specific failure records and infrastructure metadata coverage will be audited before finalising the experimental dataset. Backblaze's overall fleet statistics should not be interpreted as SSD-only statistics.

Expected Outcomes

- **Failure Pattern Analysis:** Identify failure concentrations across SSD models and infrastructure groups.
- **Maintenance Prioritisation:** Generate ranked maintenance lists based on historical failure distributions.
- **Budget-Based Evaluation:** Compare random, model-aware, infrastructure-aware and multi-factor selection strategies under different maintenance budgets.
- **Performance Assessment:** Evaluate strategies using failure coverage, selection precision and improvement over random selection.
- **Decision Support Dashboard:** Visualise failure trends, maintenance priorities, strategy comparisons and resource utilisation.

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Thank You

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Methodology

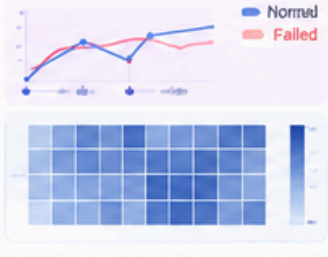
01 Data Preparation

- Load and preprocess Alibaba SSD dataset
- Clean missing values and outliers
- Merge failure, location and SMART information
- Feature engineering



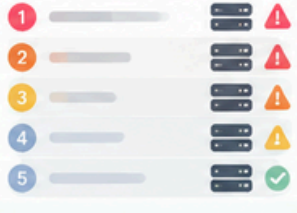
02 Failure Analysis

- Analyze failure proportions by model, rack and node
- Explore SMART attribute trends
- Identify high-risk groups and patterns



03 Maintenance Strategies

- Random selection (baseline)
- Model-based ranking (risk scores)
- Rack/node concentration-based prioritization
- Define maintenance sets under constraints



04 Budget Simulation

- Select top drives under a fixed budget
- Simulate different maintenance policies
- Compare coverage of historically failed drives
- Evaluate cost–risk trade-offs



05 Visualization & Tool

- Ranked drive list with risk scores
- High-risk racks/nodes visualization
- Comparison across maintenance strategies
- Interactive dashboard for analysis and decision support



Existing System

How Storage Maintenance Is Currently Challenged

- **Reactive Maintenance** : Drives are often replaced after failure or after warning signals are detected.
- **Drive-Level Monitoring** : SMART attributes and health indicators are commonly used to assess drive condition.
- **Uneven Failure Distribution** : Failures can vary across device models and infrastructure groups.
- **Limited Maintenance Capacity** : Maintenance teams must decide which drives to inspect or replace first.

Our Focus

Rather than developing another failure prediction model, we investigate how historical failure information can support more effective maintenance allocation.

Identification of problem

- Increasing Drive Failures: Large-scale storage systems face inevitable drive degradation and failures.
- Limited Resources: Inspecting and replacing every drive is costly and impractical.
- Insufficient Early Warning: Traditional monitoring may not accurately identify all impending failures.
- Lack of Prioritization: Maintenance teams need to distinguish high-risk drives from healthy ones.
- Need for Predictive Maintenance: Historical SMART telemetry and machine learning can support risk-based maintenance decisions.

Identified Problem: Develop a data-driven system to predict drive failure risk and prioritize maintenance under limited resources.

What this project solves

- Early Failure Detection: Identifies drives showing patterns associated with potential failure.
- Risk-Based Prioritization: Ranks drives based on their estimated failure risk.
- Reduced Downtime: Helps prevent unexpected storage failures and service disruptions.
- Cost Optimization: Reduces unnecessary inspections and premature drive replacements.
- Efficient Resource Allocation: Helps maintenance teams focus on the most critical drives first.

Core Solution: A machine-learning-based predictive maintenance system that converts historical drive data into actionable maintenance priorities.

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Dataset Overview

We use the publicly available Alibaba SSD Open Dataset, which contains real operational data collected from a large-scale production data center. The dataset provides SMART attributes, failure records, and infrastructure information, enabling analysis of drive failure patterns and maintenance strategies.



Dataset Files Used

ssd_failure_tag.csv Historical failure records for SSDs	~14,985 drives Failure labels
location_info_of_ssd.csv Rack, node and disk location information	~14,985 drives Infrastructure data
smart_log_20191231.csv SMART attributes (sample)	~5.9M records Time-series logs

Dataset Schema (Key Features)

Category	Example Fields
Failure Information	disk_id, failure, failure_time, first_failure_time
Location Information	disk_id, model, manufacturer, server_id, rack_id, node_id, slot_id
SMART Attributes	timestamp, disk_id, smart_id_smart_value, normalized_value, raw_value
Operational Metadata	model, capacity, manufacturer, application_type

Key Information in Dataset

Time Period January 2018 – December 2019	2 Years
Total Drives ~14,985 SSDs	Large Scale
Failure Tickets Actual failure records with timestamps	Real-world
Drive Models & Manufacturers Multiple SSD models and vendors (e.g., Intel, Samsung, HGST, etc.)	Diverse
Infrastructure Information Server / rack / node / slot location	Hierarchical
SMART Attributes Multiple SMART metrics for each drive (sampled over time)	Time-series
Application Type Different usage types in production environment	Realistic

Dataset Characteristics

~14,985 Total SSDs Unique drives in dataset	~5.9 Million SMART Records Time-series entries	2 Years Data Period Jan 2018 – Dec 2019	Multiple Drive Models Various manufacturers

Dataset Timeline

2018 — Jan 2018: Data collection starts

2019 — Dec 2019: Data collection ends

Real-World Data Center Environment
Collected from a large-scale production data center with real operational workloads.

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System Architecture

